

Briefly on Grammatical Representation

What is Grammar?

- Several definitions:
 - The rules and principles that guide the structure of sentences in language
 - Regularities in the mapping between phonological form and semantic form
 - The way a language speaker organizes her experience with language, allowing her to make generalizations

Why do we need grammar?

- Grammatical analysis in NLP involves breaking the text into simpler parts and categorizing them
 - New sentences are the rule, rather than the exception (unlike new words)
- Two main motivations for grammar:
 - **Structural Analysis:** decomposing complex units into simpler sub-structures can assist learning and generalization
 - **Semantic Information:** grammatical structure often reflects semantic structure and distinctions

Text Strings are not Informative Enough, even with POS Tags

Q: Where were the crates taken?

The terrorists took the crates they found **to** the hangar

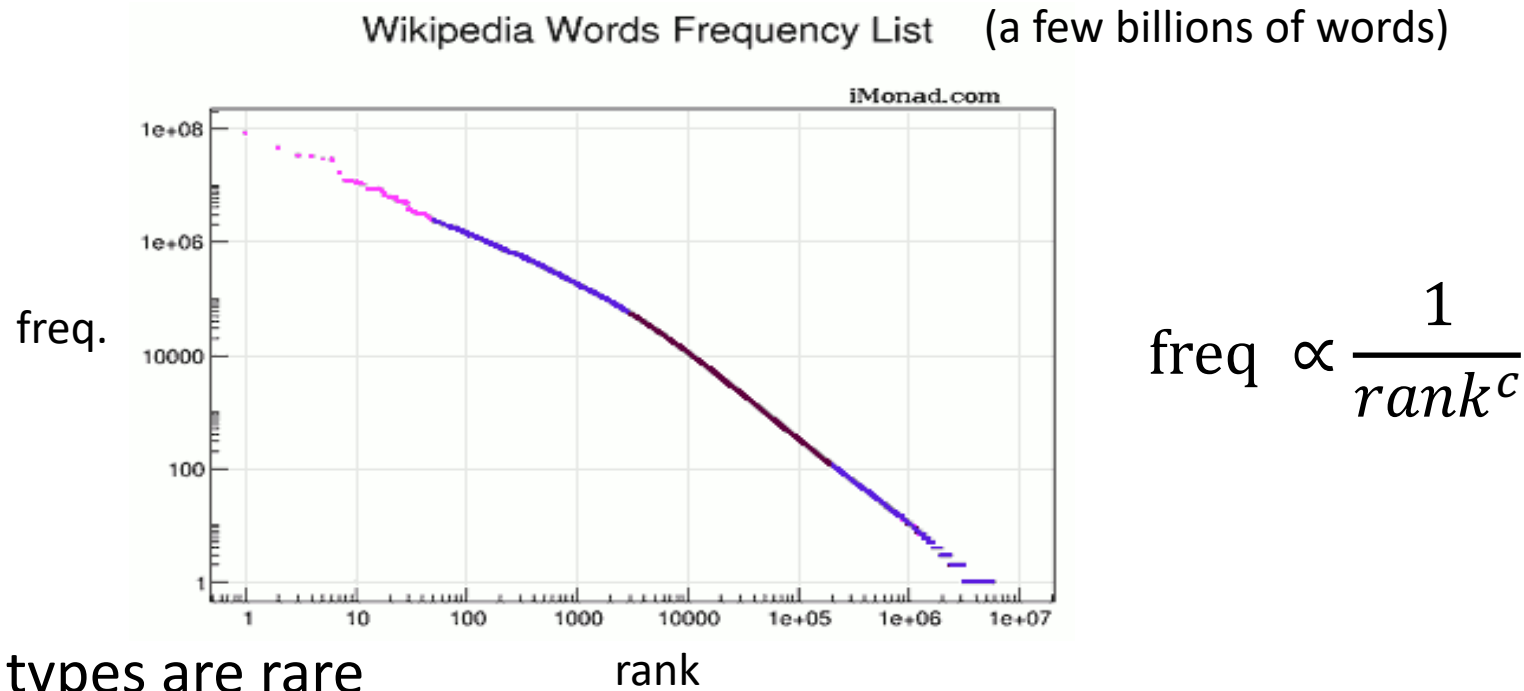
A: to the hangar

The terrorists took the crates they found **in** the hangar

A: *not* to the hangar

The Importance of Categorization

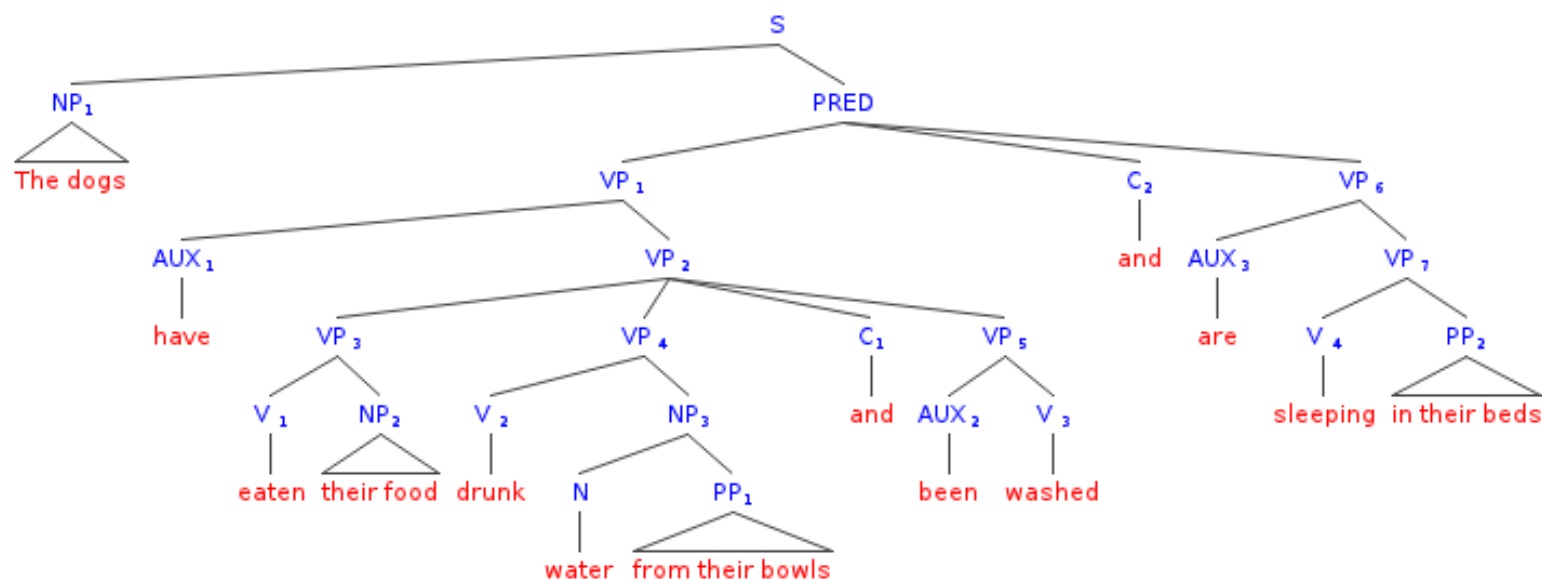
- Reminder: linguistic phenomena tend to have a power-law distribution



- Most word types are rare
- Many word instances are rare

The Importance of Categorization

- This means that for most types, we don't have meaningful statistics
 - Many types don't appear at all
- Categorization allows information to propagate from one type to another



Overview of Grammar: Words

Basic Linguistic Theory, Vol. 1-2, Dixon (2010)

- **Word:** surprisingly difficult to define
 - **Practical:** anything that is between two spaces
 - But: what about spoken language?
 - what about phrases whose spaces disappeared?
("kick start", "kickstart" or "kick-start"?)
 - What about languages that don't have spaces?
(e.g., Mandarin Chinese)



Overview of Grammar: Words

- Sapir (1921): “one of the smallest, completely satisfying bits of isolated meaning into which the sentence resolves itself”
- Zirmunskij (1966): “the word is the most concise unit of language, which is independent in meaning and form”
- Bloomfield (1933): “a word, then, is a free form which does not consist entirely of (two or more) lesser free forms; in brief, a word is a *minimum free form*”
 - “free form” is interpreted to mean words that can occur on their own, with a given meaning
 - But: many words (“the”, “my”, “and”) cannot occur on their own

Overview of Grammar: Words

- We will distinguish between:
 - **Phonological words:** defined by phonological criteria, primarily: syllable structure, pause phenomena, prosodic features (such as stress or tone), phonological word boundary rules
 - **Grammatical words:** have a conventionalized meaning, can undergo morphological processes (e.g., affixation, reduplication, vowel shift)

Morphology

- Morphology is the part of grammar that addresses the inner structure of words
- While English has a rather impoverished morphology, other languages have a much richer one (e.g., Hebrew)
- Morphological analysis is the decomposition of words into their sub-parts and the structure that combines them
- In NLP, morphological analysis minimally involves segmentation into tokens (the minimal units of analysis)
 - There are many tokenization schemes

Morphological Analysis in NLP

- Basic tokenization: (mostly separating out clitics)
 - In English, mostly involves separating out apostrophes:
 - isn't → is+n't
 - let's → let+'s
 - Dina's --> Dina+'s
 - ספר, ה', בית, ל' will turn into ['ל', 'בית', 'ה', 'ספר']
 - Auf'm → auf 'm (German: auf+dem)
- Lemmatization: mapping to the base form
 - dogs → dog
 - כלבים → כלב
 - כלב → كلب (lit. dogs → dog)
 - takes → take
 - נתן → יתנו

Segmentation into Morphemes

- Some languages combine words into compounds or other morphologically complex structures
 - *renewal* → *re + new + al*
 - *German: nachgedacht* → *nach + ge__t + dach* (lemma: *denken*)
 - *מטבחונים* → *ים + ון + מטבח*

Tokenization of a Hebrew Sentence

1	כתובת	כתובת			
2-3	בארמנית	–			
2	ב	ב			
3	ארמנית	ארמנית	19-20	בציפורים	–
4-5	המתנוססת	–	19	ב	ב
4	ה	ה	20	ציפורים	ציפור
5	מתנוססת	התנוסס	21	שונות	שונה
6-7	בראש	–	22	,	,
6	ב	ב	23-24	בכלים	–
7	ראש	ראש	23	ב	ב
8-9	הפסיפס	–	24	כלים	כלי
8	ה	ה	25-27	ובפירות	–
9	פסיפס	פסיפס	25	ו	ו
10	,	,	26	ב	ב
11-12	וכן	–	27	פירות	פרי
11	ו	ו	28	,	,
12	כן	כן	29	מלמדים	לימד
13-14	סגנונו	–	30	כי	כי
13	סגנון	סגנון	31	נעשה	נעשה
14	ו	הוא	32	בידי	בידי
15-16	האמנותי	–	33	אמנים	אמן
15	ה	ה	34	ארמנים	ארמן
16	אמנותי	אמנותי	35	.	.
17-18	העשיר	–			
17	ה	ה			
18	עשיר	עשיר			

Basic Syntactic Units: Clauses

- **Clause:**
 - Semantically: “the description of some activity, state or property”
- Simple clauses have a predicate, core arguments (usually subjects and objects), and non-core arguments (e.g., location, manner, instrument)
 - “John **kicked** the ball”
 - “The door **opened**”
 - “The door silently **opened**”
 - “John **opened** the door with a hammer”
 - “John is **tall**”

Inter-Clause Relations

- Three main inter-relations between clauses within a sentence: complement clauses, relative clauses, linked clauses, coordination
- Complement clauses:
 - Clauses that serve as an argument in another clause
 - “John said **he will kick the ball**”
 - “Mary convinced Paul **to join them**”
- Coordination:
 - “**I got home** and **took a shower**”
- Relative clauses:
 - Clauses that modify an argument of another clause
 - “John, **who has studied the language for years**, speaks German well”
 - “Dogs **that bark** never bite”
- Linked clauses:
 - After **he had studied it for years**, **John could speak German well**
 - Although **he had studied it for years**, **John could speak German well**

Basic Units: Noun Phrases

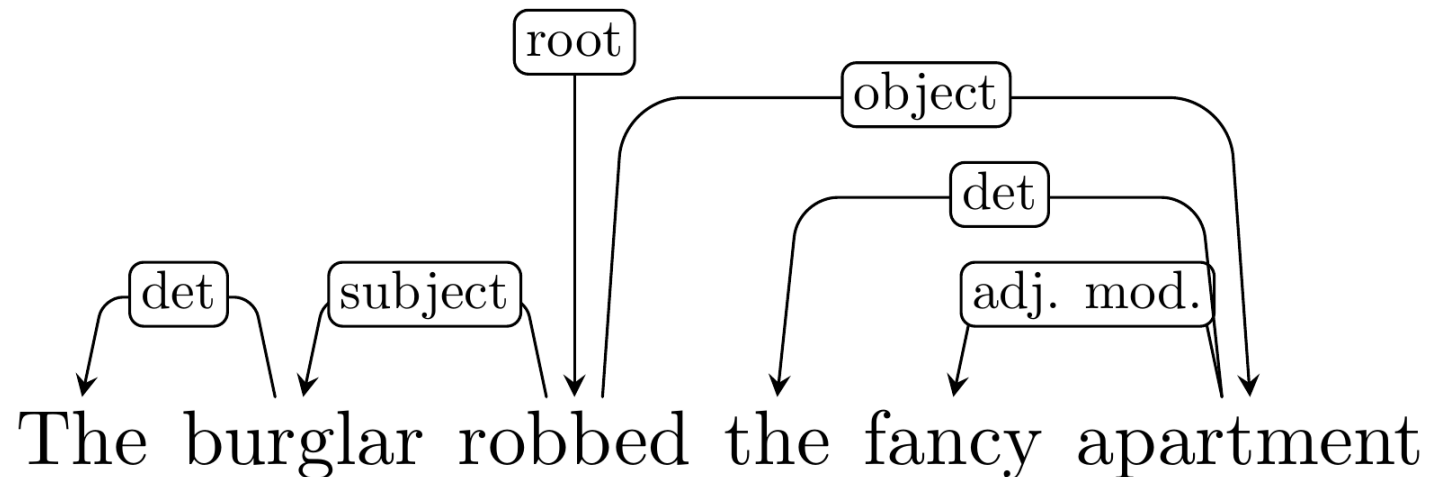
- **Noun phrase:** an argument of a clause is generally a noun phrase. Two major classes:
 - Proper nouns or names (of people, places, organizations etc.)
 - Common nouns, possibly along with some modifiers
 - Possible modifiers: adjectives (e.g., fast, thin), relative clauses, cardinals or ordinals (e.g., one, first), modifying nouns (e.g., shipping company)
 - Noun phrases generally have a *head*, a word that determines the semantic type of the whole phrase, and is an essential component of its meaning
 - Shipping **company**
 - Fast **runner**
 - The **man** who wasn't there

Basic Units: Syntactic Heads

- The notion of a *head* is used in a wider context than noun phrases
- The verb is considered the head of a clause:
 - Arguments and modifiers are considered its *dependents*
 - **John** ran **home** (“ran” is the head of the clause, “John” and “home” are its dependents)
 - **Bill stupidly** answered **the officer’s question** (“answered” is the head, “Bill”, “stupidly” and “the officer’s question” are its dependents)
- Some schemes view all syntactic relations as head-dependent relations (*dependency schemes*)

Dependency Parses

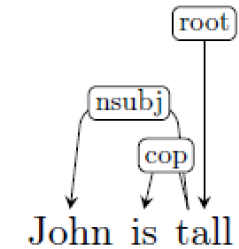
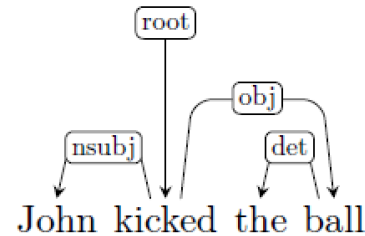
- Represent Syntax as head-dependent relations between words
 - Nodes are words, edges are relations
 - The arguments and modifiers of a word are its children
 - Edge labels mark the type of relation
 - The verb of the main clause is the root



Example Annotations in *Universal Dependencies* Format

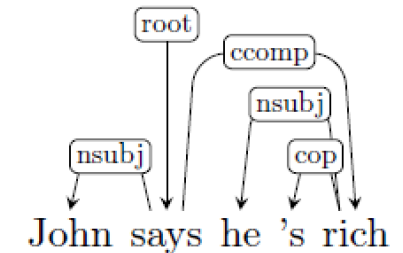
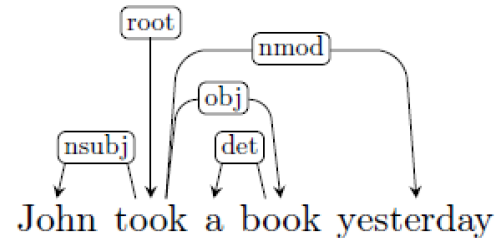
- Transitive clauses:

- John kicked the ball
- John took a book yesterday



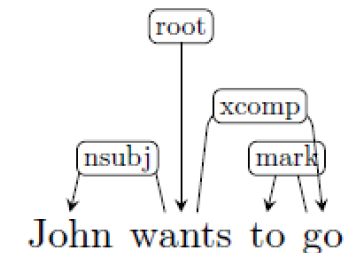
- Copula clauses:

- John is tall



- Complement clauses:

- John says he's rich
- John wants to go

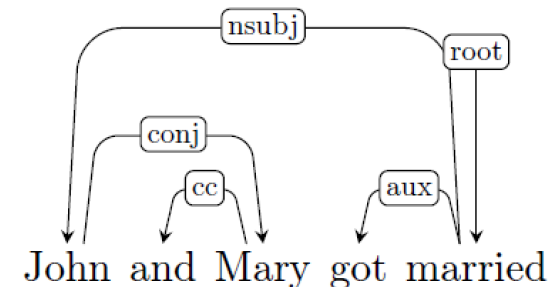
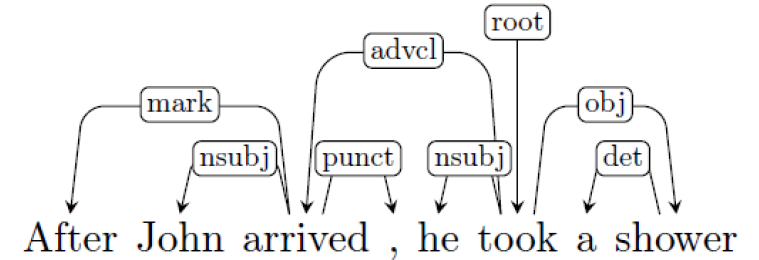
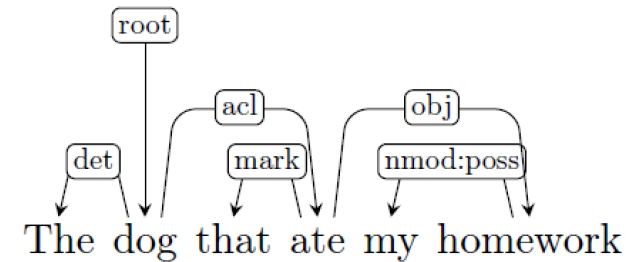


<http://universaldependencies.org/>

<http://universaldependencies.org/u/dep/index.html>

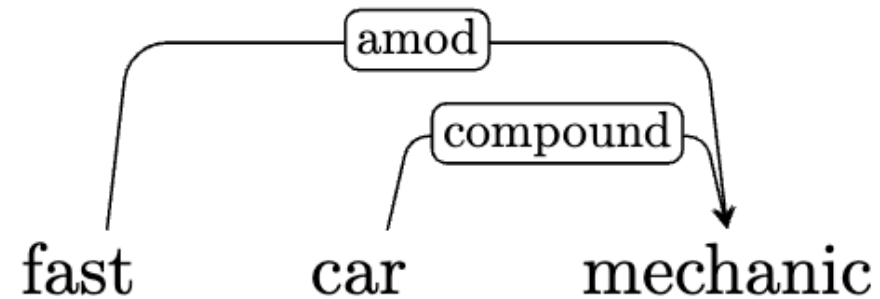
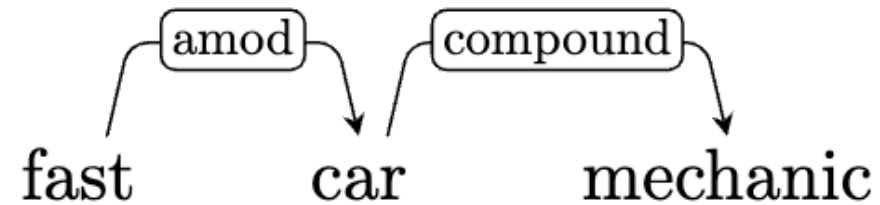
Example Annotations in *Universal Dependencies* Format

- Relative Clauses:
 - The dog that ate my homework
- Subordinate clauses:
 - After John came home, he took a shower
- Coordination:
 - John and Mary got married

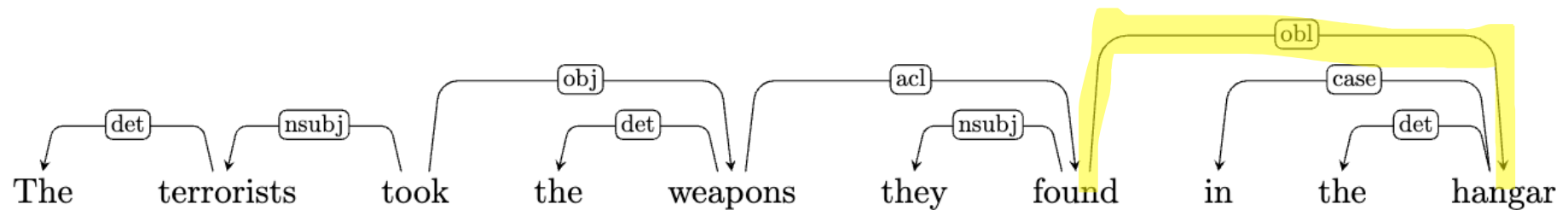
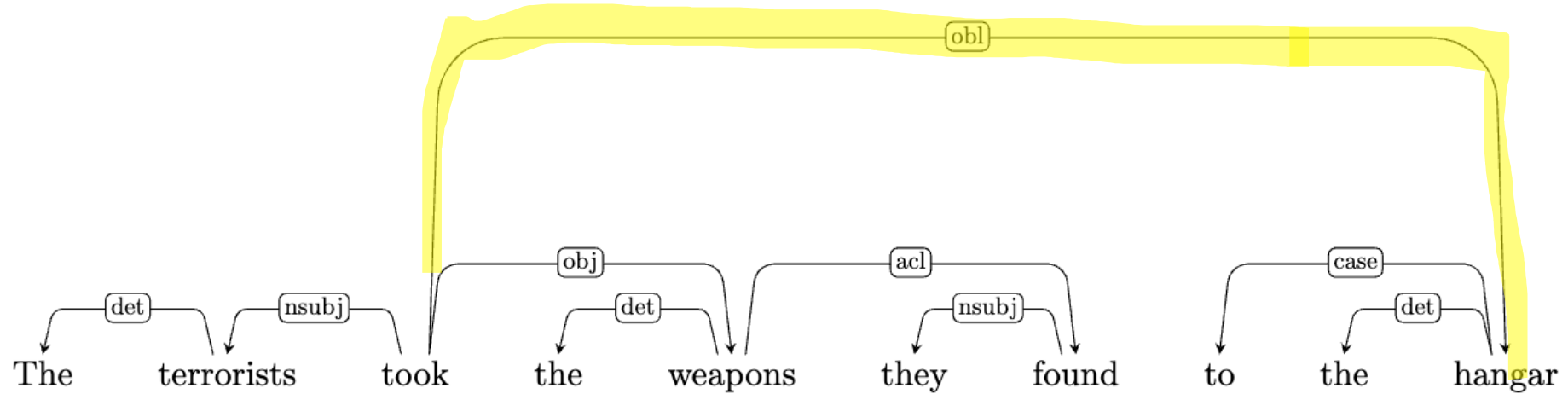


Information from Syntactic Structure

- Syntactic disambiguation: what does the adjective modify?



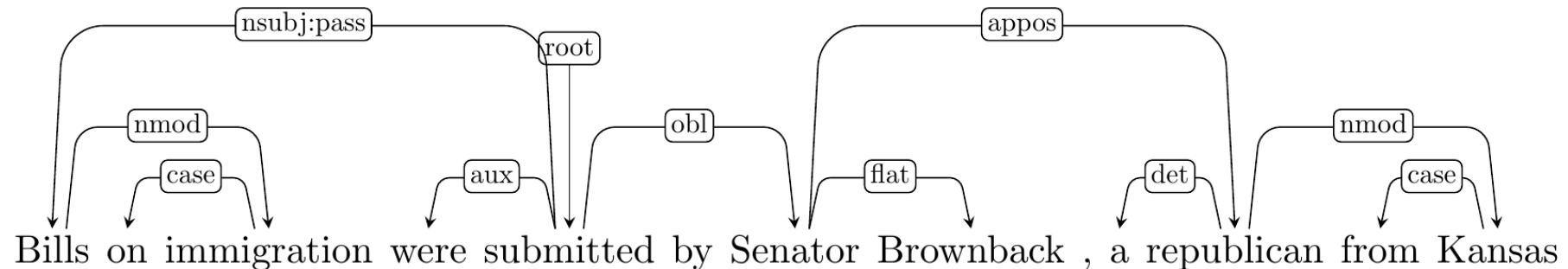
Information from Syntactic Structure



A Proxy to Semantics

- Syntactic structure provides valuable semantic information that can serve downstream applications, such as paraphrasing, machine translation, summarization etc.

Bills on immigration were submitted by
Senator Brownback, a republican from Kansas



A Proxy to Semantics

- Allows answering questions:
 - What was submitted?
 - Who submitted them?
 - What is Senator Brownback?
 - What does “a republication from Kansas refers to?”

